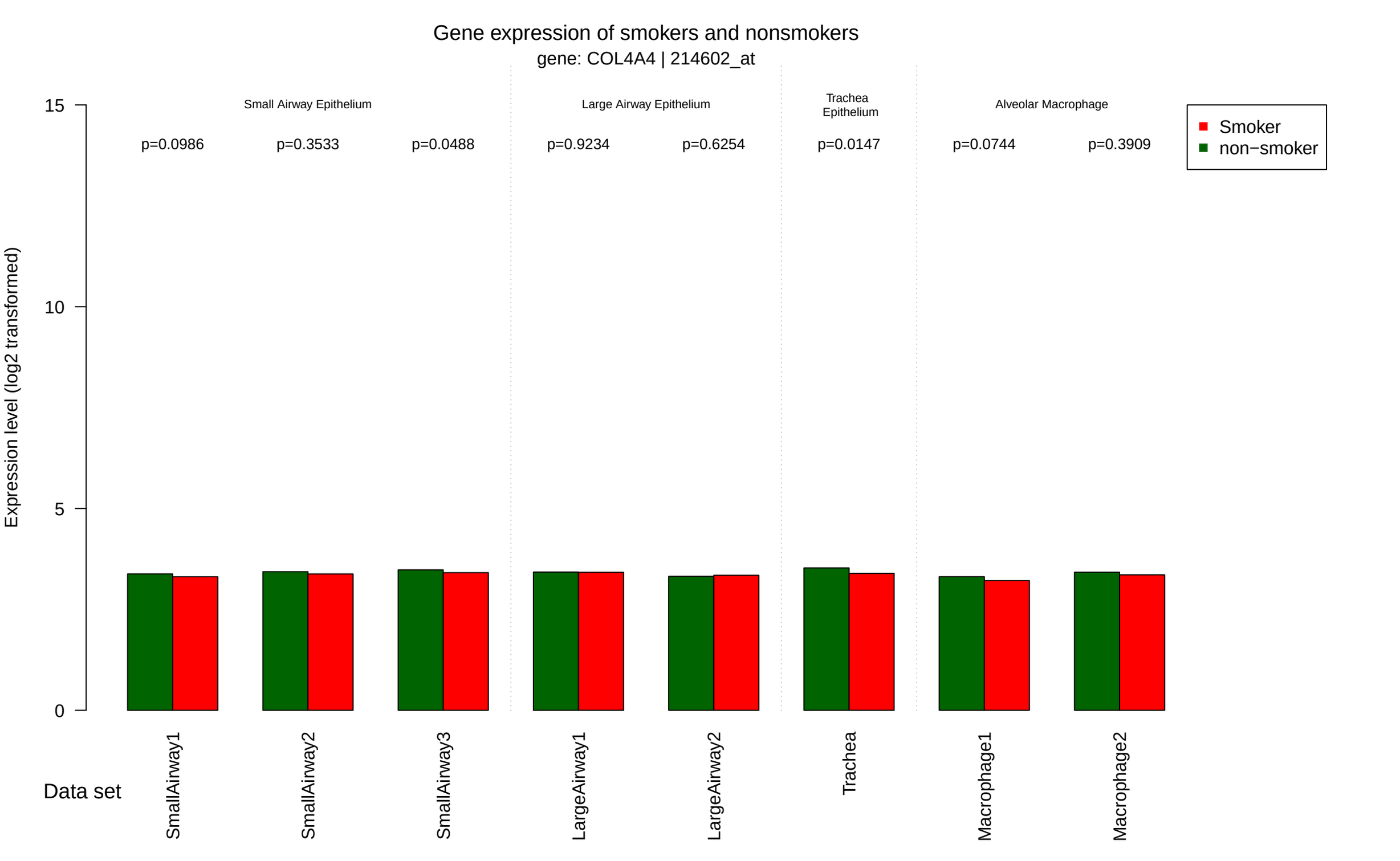
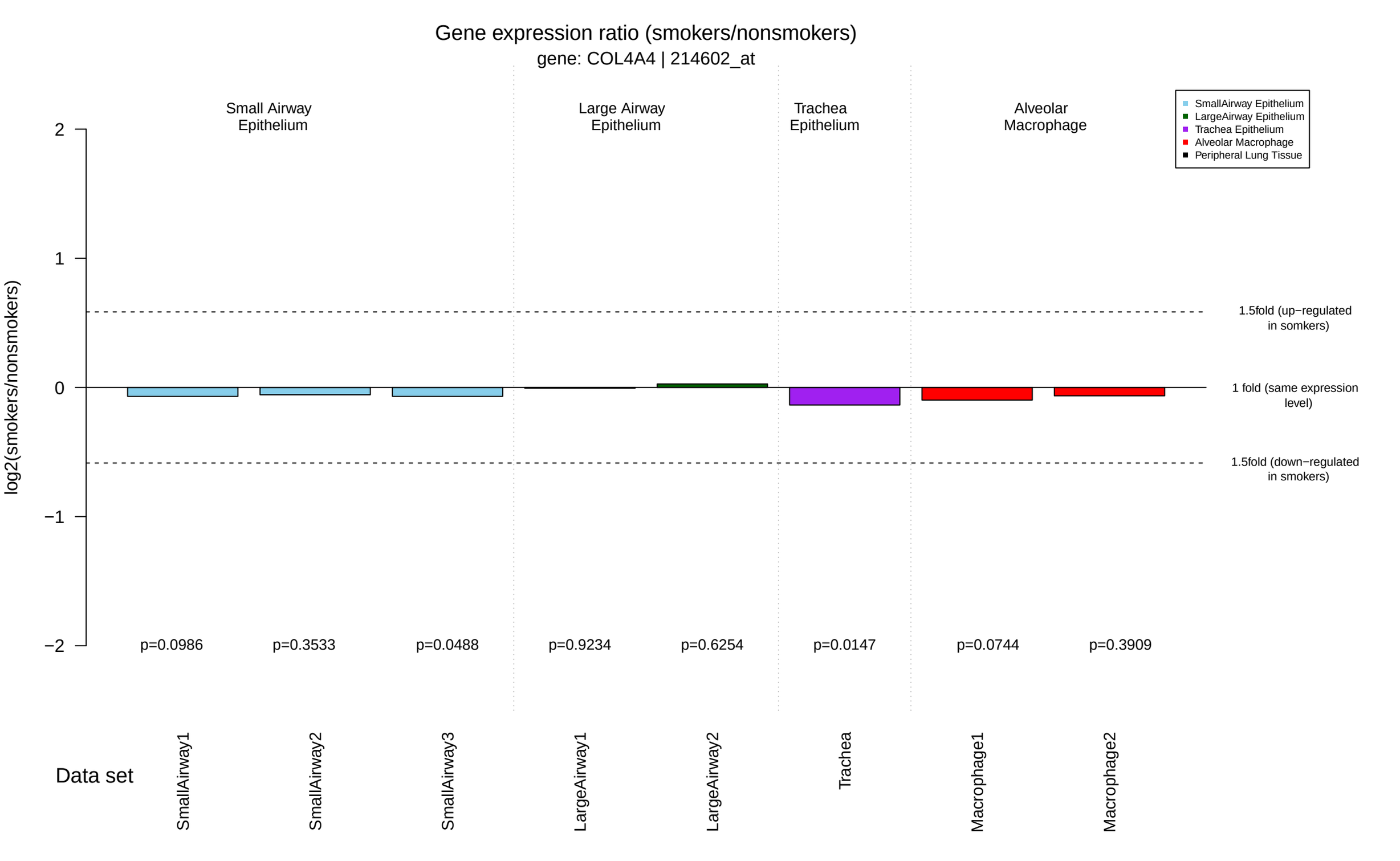
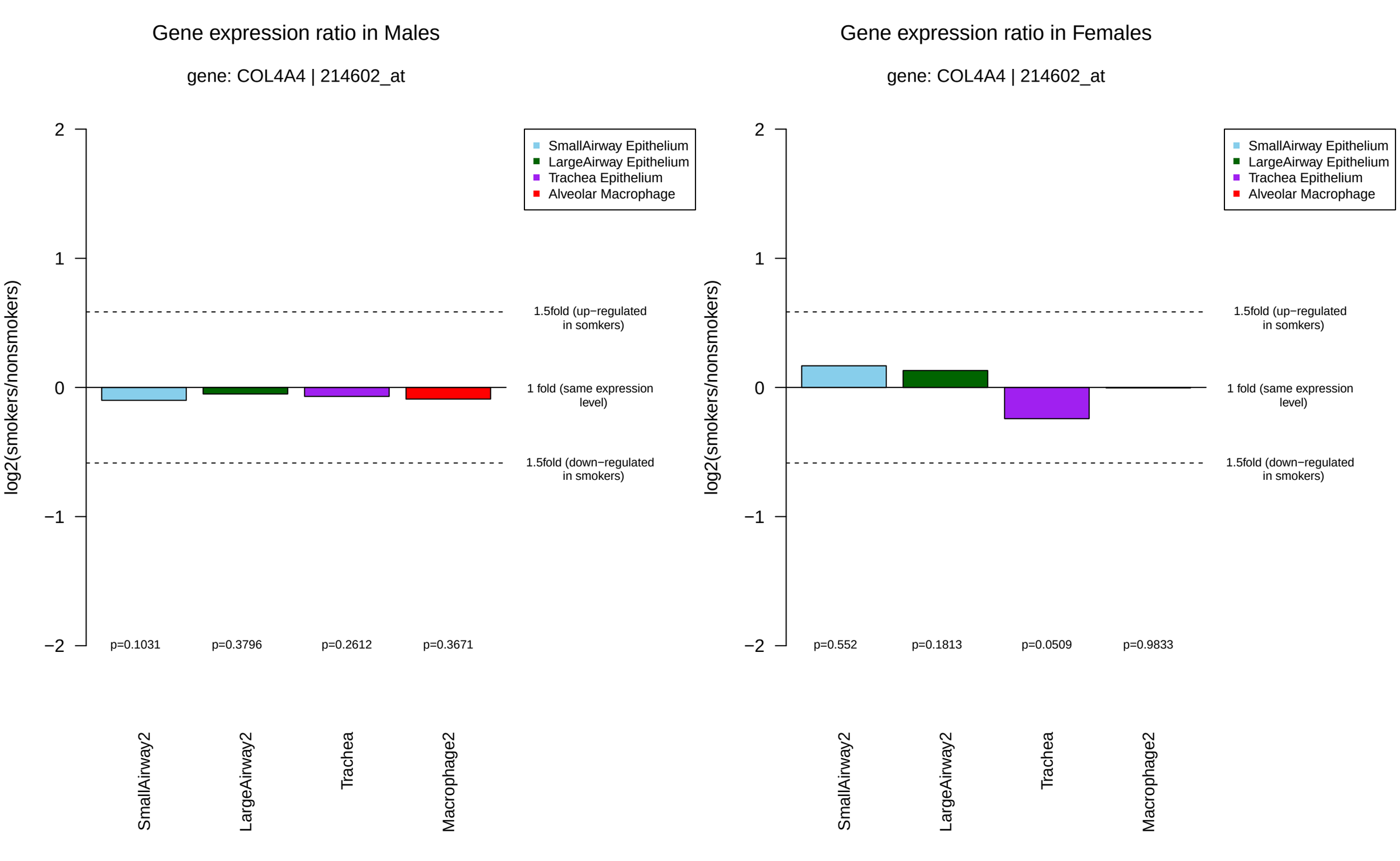
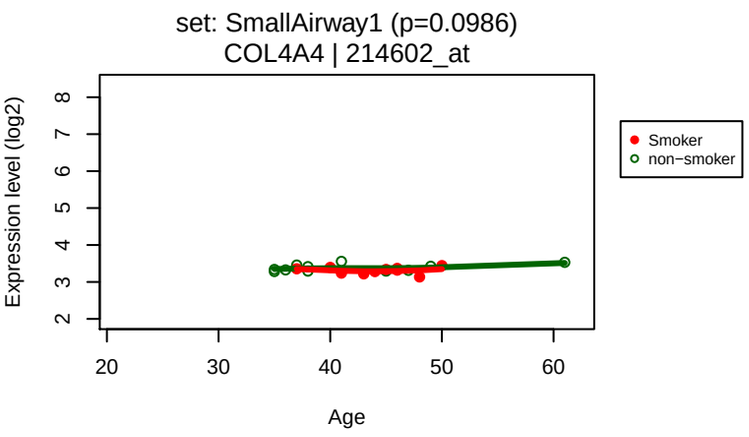
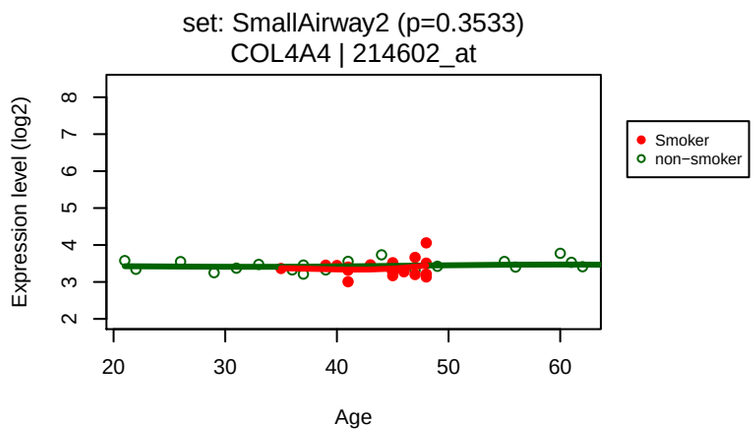
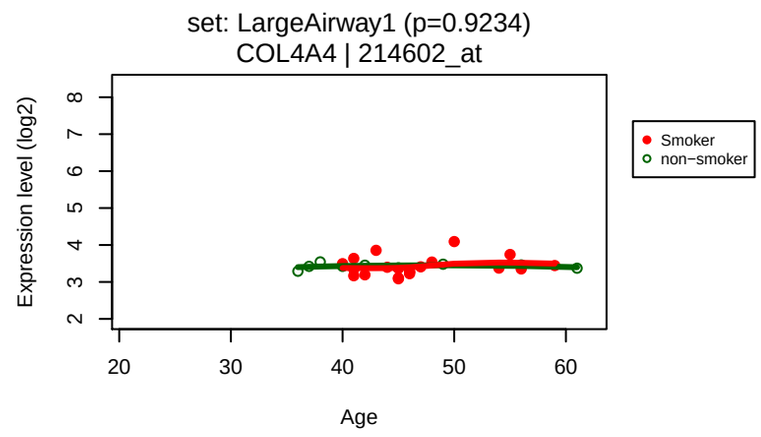

Gene expression of smokers and nonsmokers
gene: COL4A4 | 214602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0986
p=0.3533
p=0.0488
p=0.9234
p=0.6254
p=0.0147
p=0.0744
p=0.3909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

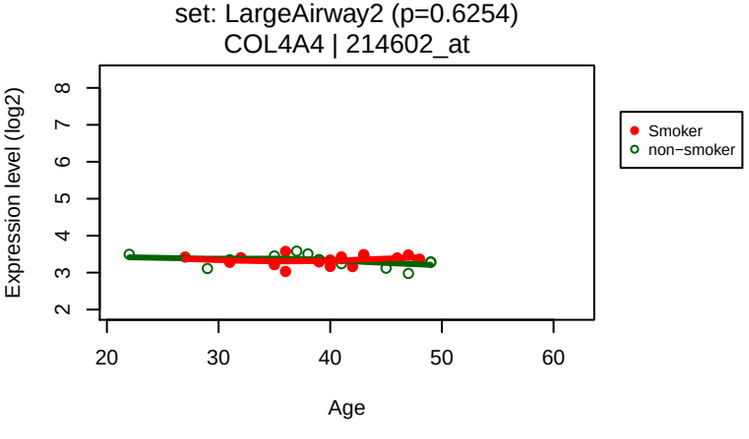

Gene expression ratio (smokers/nonsmokers)
gene: COL4A4 | 214602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0986
p=0.3533
p=0.0488
p=0.9234
p=0.6254
p=0.0147
p=0.0744
p=0.3909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

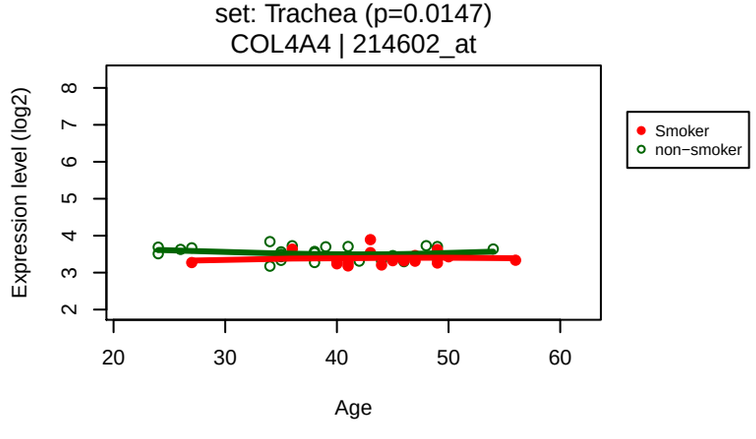

Gene expression ratio in Males
gene: COL4A4 | 214602_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1031
p=0.3796
p=0.2612
p=0.3671
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A4 | 214602_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.552
p=0.1813
p=0.0509
p=0.9833
SmallAirway2
LargeAirway2
Trachea
Macrophage2

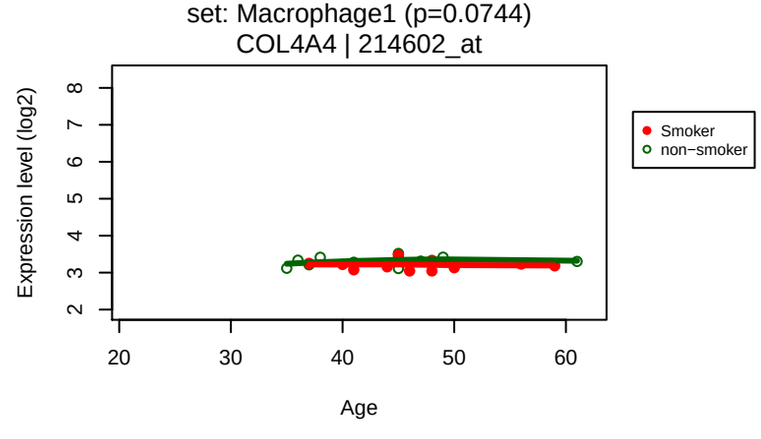

set: SmallAirway1 (p=0.0986)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker

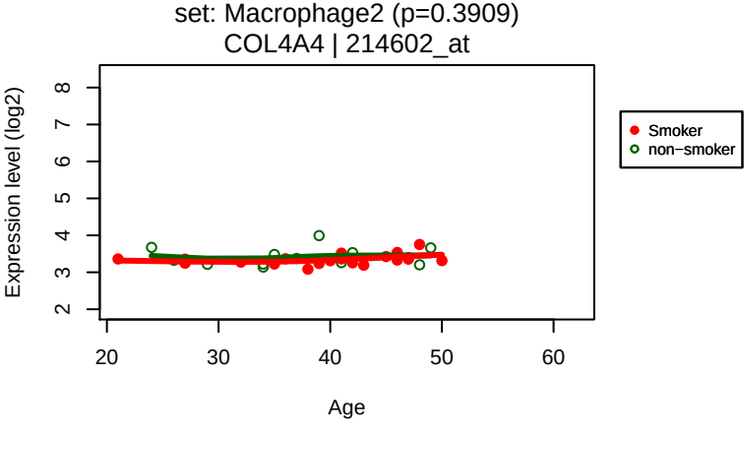

set: SmallAirway2 (p=0.3533)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker

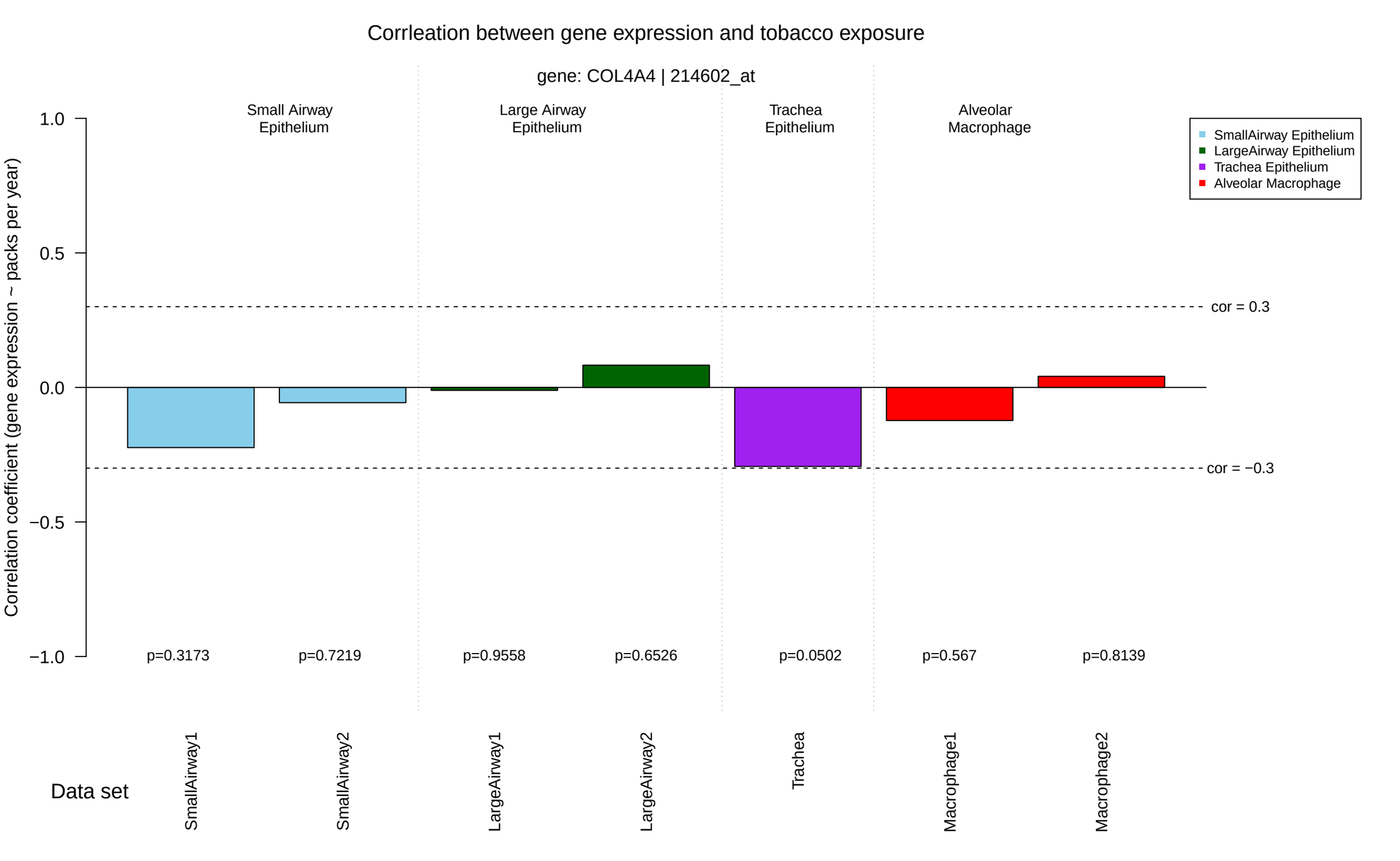

set: LargeAirway1 (p=0.9234)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6254)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0147)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0744)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3909)
COL4A4 | 214602_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A4 | 214602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3173
p=0.7219
p=0.9558
p=0.6526
p=0.0502
p=0.567
p=0.8139
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2